PRANAV DHIRAN

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RESEARCH OVERVIEW

Researcher-engineer working on small language models, agentic systems, and verifiable LLM systems. At AIISC (University of South Carolina) I own pre-training for a 140M-parameter agriculture SLM, work on GRPO-based RL fine-tuning, and built India-Agri-KG, a 784-district agricultural knowledge graph. First-author paper at NeurIPS 2026 AI4Good.

RESEARCH EXPERIENCE

Research Intern — AIISC, University of South Carolina Apr 2026 – Present

- Own the pre-training pipeline for a 5-person team building an India-focused agriculture SLM (data to evals).
- Designed and pre-trained a **140M-parameter Qwen3-style transformer** (GQA, RoPE, SwiGLU) on a 6.5 GB corpus. Neurosymbolic layer: AGROVOC entity-injected BPE tokenizer, causal KG verification, Triple Transformer Encoder. GRPO post-training.
- Built **India-Agri-KG** (784 districts, 26 domains, 14,478 entities, 38,590 relations) with four LLM-directed collection agents and a deterministic non-LLM verification layer enforcing source traceability on every triple.

PUBLICATIONS

Navigate, Don't Generate: Making AI-Mediated Access to Public Agricultural Data Citizen-Auditable

First author · NeurIPS 2026 AI4Good Workshop OpenReview

- LLM agents propose triples from government sources; a deterministic 5-stage non-LLM gate alone accepts or rejects each, so every triple carries a source URL and access date.
- Showed the Soil Health Card fertilizer API is input-invariant (20/20 probes), recovering 4,618 records; compared against 6 deployed agri-advisory systems and 4 KG-construction baselines.

ENGINEERING EXPERIENCE

Lead Developer — Charak Aug 2026 – Oct 2026

- Led development of the apps and website, scaling to **500+ users**; owned technical design, releases, and production deployment.

LFX Mentee — Hyperledger Cello, Linux Foundation Jun 2026 – Sep 2026

- Built an **AI operations copilot for Hyperledger Fabric**: an LLM tool-calling agent mapping natural-language queries to Cello REST APIs, with a Django/DRF backend and SSE streaming to the React dashboard.
- Built a Fabric node-logs API with bounded log retrieval and Docker error handling, and a multi-party channel invitation workflow end to end (DB models, REST APIs, Fabric signing, UI). **21 PRs** across features, debugging, and CI fixes.

PROJECTS

GNU Radio MCP Server Python · FastMCP · ZMQ · XML-RPC PyPI | GitHub

- Published MCP server exposing **13 tools** (flowgraph lifecycle, frequency/gain control, signal monitoring) to LLMs, with Pydantic v2 validation and ZMQ IQ streaming; used with HackRF One and RTL-SDR.

Medaura — Agentic Pharmacy System FastAPI · LangChain · ChromaDB · Langfuse Live | GitHub

- 5-agent pharmacy system (order, safety, forecast, procurement) with ChromaDB RAG and Langfuse tracing; live in 4 languages.

RF Watch — RF Spectrum Monitor Python · GNU Radio · HackRF One GitHub

- Real-time spectrum analyzer with FFT features and an ML classifier for passive transmitter detection (SIH 2025).

TEACHING, SERVICE & AWARDS

Lecturer & Evaluator, IAIRO SLM++ Bootcamp 2026 (AIISC): Vanilla GPT-2 Architecture · Scaling Laws & Cost Accounting

Reviewer, AI4Good Workshop @ NeurIPS 2026 (recognition letter)

Awards, International Finalist (Top 6) UWA Hack 2026 · National Finalist SIH 2024 & 2025

TECHNICAL SKILLS

Languages & ML	Python, PyTorch, Hugging Face Transformers, TensorFlow, Go
LLM / SLM	Transformer Pre-training, BPE Tokenization, Scaling Laws, Distributed Training, Neurosymbolic AI, Knowledge Graphs
Post-training	GRPO, RLHF, Instruction Tuning, LoRA, PEFT, Quantization (INT4/INT8), TRL, Unsloth
Agentic	Tool Use, Function Calling, Model Context Protocol (MCP), Multi-Agent Systems, LangChain, LangGraph, RAG, ChromaDB, FAISS, Langfuse
Tools	Git, Jupyter, REST APIs, gRPC, Docker, Kubernetes (basic)

EDUCATION

B.Tech — Electronics & Telecommunication Engineering (Minor in IT), SGGS IE&T, Nanded 2023 – 2027

Courses: Linear Algebra & Calculus, Probability & Statistics, Discrete Math, Data Structures & Algorithms, DBMS, Signals & Systems, DSP